

TRAGIC AFTERMATH
Relief personnel carry the body of a victim, Jul. 30

AFP

A 2022 study in the *International Journal of Environmental Research and Public Health* showed that 62 per cent of the forests in Wayanad had disappeared between 1950 and 2018 while plantation cover rose by 1,800 per cent. Climate change, the unclaimed child of similar worldwide devastation of forests, has now come back to haunt the scene of the crime, with extreme weather phenomena aggravating things precisely in vulnerable places like the Western Ghats, one of the eight "hottest hotspots" of biological diversity in the world.

Scientists at the Advanced Centre for Atmospheric Radar Research at the Cochin University of Science and Technology (CUSAT) have found that rainfall over India's west coast is becoming more convective. "The warming of the Arabian Sea led to the formation of deep cloud systems, leading to heavy rains in short periods in Kerala which in turn increases the possibility of landslides," says the centre's director S. Abhilash. The warming sea causes the air above to become thermodynamically unstable. "This atmospheric instability is linked to climate change. Earlier, this kind of rainfall was more common in the northern Konkan belt, north of Mangaluru," says Abhilash.

The 2022 study suggested that warnings on torrential rain and related disasters cannot be ignored. The Hume Centre for Ecology and Wildlife Biology, which collects rainfall data from over 200 locations in Wayanad district, had apparently alerted the administration about the possibility of landslides in Mundakkai and surrounding areas 16 hours before the disaster, at 9 am on July 29. The centre's data had shown that Puthumala, the nearest rainfall gauge to Mundakkai, recorded 200 mm on July 28 and another 130 mm overnight.

"A landslide can be triggered by about 600 mm of rain. So, we promptly issued an alert that further rainfall could be critical," says Hume Centre director C.K. Vishnudas. The area had received 572 mm of rain in 48 hours since July 28. In 2020, a Hume Centre warning of an impending landslide in Mundakkai had led to the successful relocation of people, preventing casualties. This time, too, the district administration did some evacuation but, as it turned out, not nearly enough.

Equally embarrassing, for both the state and the Centre, is that an advanced early warning system, inaugurated in July, 'failed' to alert the administration. The system, which uses artificial intelligence, reportedly did not forecast the devastating landslide. Union minister of coal and mines G. Kishan Reddy inaugurated the National Landslide Forecasting Centre (NLFC) at the Geological Survey of India headquarters in Kolkata on July 19, along with the unit in Wayanad. The NLFC, aimed at landslide hazard mitigation, is expected to provide early information to the local administration, update landslide inventories and integrate real-time rainfall and slope instability data.

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Indeed, whether advanced warnings were given or not has become a bone of contention, with Union home minister Amit Shah claiming it was done seven days earlier, on July 23, and that nine NDRF (National Disaster Response Force) teams were sent to the state as a contingency response. Kerala chief minister Pinarayi Vijayan countered this, saying they had got no such warnings, and cited the meteorological department's orange alert notifications for June 29-30 to bolster his case, and asking Shah to "stop playing politics".

The blame game aside, apprehensions abound in Wayanad and other landslide-vulnerable areas about possible evacuation (and relocation) to prevent a recurrence of the tragedy. The human dilemma about the challenges of relocation without adequate compensation is one the state will have to deal with sooner or later. "We discuss community security only when tragedies happen and ignore the basics of handling disaster management," laments biologist and conservationist P.E. Easa. "The reality is that Kerala has to cope with high population density and limited spaces. So, effective disaster management practices are critical." This is something everyone knows. Whether they choose to do something about it is another matter. ■